

Omid Ostovari

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Educations

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| MS.c University of Tehran , Artificial Intelligence and Robotics
<ul style="list-style-type: none"> Thesis: Calcium Imaging Denoising Using Deep Self-Supervised Models Advisor: Prof. Vahabie GPA: 3.7/4.0 (Last two semesters: 4.0/4.0) Coursework: Cognitive Science (4.0/4.0), Deep Neural Networks (4.0/4.0), Machine Learning (4.0/4.0) | Sep 2022 – Sep 2025 |
| BS.c University of Yazd , Computer Engineering
<ul style="list-style-type: none"> Thesis: Decision Support System for Stock Trading Advisor: Prof. Derhami Coursework: Software engineering (4.0/4.0), Project Management (4.0/4.0) | Sep 2016 – Sep 2021 |

Research Interests

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| • Volumetric deep learning | • Computational neuroimaging | • Self-supervised learning |
| • Neural decoding | • 2-Photon calcium imaging | • Neural dynamics |

Publications

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| Self-supervised Calcium Imaging Denoising Using 3D UKAN
<i>Omid Ostovari¹, Abdol-Hosein Vahabie²</i> | In Progress, 2025 |
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Research Experience

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| University of Tehran , Research Assistant
<ul style="list-style-type: none"> Applied CalmAn for motion correction of two-photon calcium imaging. Designed an efficient 3D Kolmogorov–Arnold Network for volumetric denoising. Reduced memory usage by 96% and improved data throughput in PyTorch. Supervised by Prof. Vahabie | May 2023 – Dec 2025 |
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Teaching Experience

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| University of Tehran , Head Teaching Assistant
<ul style="list-style-type: none"> Course: Computer Programming, Instructor: Prof. Naderi | Fall 2025 |
| University of Tehran , Head Teaching Assistant
<ul style="list-style-type: none"> Course: Artificial Intelligence, Instructor: Prof. Naderi | Spring 2025 |
| 1st International Conference on AI , Executive Committee Member
<ul style="list-style-type: none"> Workshop: Generative Models in Computer Vision, Website | Feb. 2025 |
| Hamrah Academy , Data Science School Director
<ul style="list-style-type: none"> Course: Social Network, Instructor: Prof. Asadpour | Summer 2023 |
| Yazd University , Python Lecturer
<ul style="list-style-type: none"> Developed and taught an advanced Python course at the Students Scientific Chapter of Computer Engineering. | Fall & Spring 2019 |

Professional Experience

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| Hoko, Pishgaman Innovation Accelerator , Software Engineer | Mar. 2021 – Oct. 2021 |
| Aghazgaran Sabz, Yazd Science and Technology Park , Chief Executive Officer | Nov. 2020 – Sep. 2021 |
| Aghazgaran Sabz, Yazd Science and Technology Park , Full-Stack Developer | Jan. 2019 – Dec. 2020 |

Professional Development

Brain Dynamics Workshop Instructor: Dr. Abbaspoor, Harvard University, Harvard Brain Science Initiative	Mar. 2025
Optogenetics & Chemogenetics Workshop Instructor: Dr. Hosseinzadeh Sahafi, University of Tokyo	Aug. 2024
Calcium Imaging Workshop Instructor: Dr. Hosseinzadeh Sahafi, University of Tokyo	Jun. 2024
Spatio-temporal Modeling in Neuroscience through Machine Learning Workshop Instructor: Prof. Abbasi-Asl, University of California, San Francisco (UCSF)	Dec. 2023

Notable Projects

Denoising Diffusion Probabilistic Models for Image Generation - Implemented DDPMs on CIFAR-10 with latent diffusion tuning and FID evaluation. - Achieved improved sample fidelity compared to GAN baseline.	Fall 2023
GAN and VAE Architectures for Image Modeling - Developed stable GAN training pipeline on MNIST with mode-collapse reduction. - Designed VAE encoder-decoder for efficient latent feature learning.	Fall 2023
Multimodal Neural Imaging Dataset for Vision-Language Models - Created memory-efficient pipelines for extracting volumetric neural video-clips. - Generated metadata-based captions and domain-generalized dataloaders.	Spring 2025
SimCLR-based Contrastive Learning for Neural Image Denoising - Pretrained 3D encoder on noisy volumes using contrastive learning. - Fine-tuned for denoising with reconstruction-based loss.	Summer 2025
Underwater Image Segmentation (UNet & PSPNet) - Trained segmentation models on SUIM dataset using MobileNetV2 backbone. - Evaluated segmentation and visual performance across architectures.	Sep 2023
Artificial Neural Networks – Course Assignments - Built CNNs with regularization and training improvements. - Developed RNN/LSTM classifiers and evaluated performance.	Fall 2022
Cognitive Science – Course Assignments - Hodgkin-Huxley spiking simulation and working-memory network dynamics. - Reinforcement-learning task design and contextual feedback modeling.	Spring 2024
Natural Language Processing – Course Assignments - Built Transformer NMT models in Fairseq with BLEU evaluation. - Trained ParsBERT and RNN/GRU classifiers for sentiment analysis.	Spring 2023

Awards and Honors

Top 1%, National Entrance Exam – Master of Science in Computer Engineering	Apr. 2022
Top 2%, National Entrance Exam – Bachelor's Program	Jun. 2015

Skills

Programming Languages: Python, Java, MATLAB, C#, C++, MySQL
Python Libraries: PyTorch, Lightning, Tensorflow, Scikit-learn, Psychopy, Pandas, Numpy, CalmAn
Software & Tools: Jupyter Notebook, EEGLAB, Git, Docker, MySQL Workbench, Gephi
Languages: English (TOEFL - planned), German (A1), Persian (Native)

References

Prof. Abdol-Hossein Vahabi , School of ECE & Faculty of Psychology, University of Tehran	h.vahabie@ut.ac.ir
Prof. Ali Masoudi-Nejad , Institute of Biochemistry and Biophysics, University of Tehran	amasoudin@ut.ac.ir
Prof. Hanieh Naderi , College of Interdisciplinary Science and Technology, University of Tehran	hanieh.naderi@ut.ac.ir